

image Analysis and Healthcare Navigation

Guided By

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PRESENTED BY — BATCH 2023120304 - SECTION C (23ITMNP-C06)

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- 250,000+ preventable deaths occur annually globally due to diagnostic delay and medication errors (BMJ, WHO).

Introduction

• 40% of clinical decisions depend on diagnostic reports (blood panels, radiographs, ECGs) that patients struggle to interpret.

- Developing nations face a severe specialist shortage (India: 1 doctor per 1,457 citizens vs 1:1,000 WHO norm).
- Unstructured medical images (handwritten prescriptions, skin lesions, noisy ECGs) lack automated consumer-facing triage.

Domain: Image Processing & Healthcare Artificial Intelligence

- Combines multimodal computer vision (Pillow CLAHE, Lanczos scaling, grid filtering) with Google Agent Development Kit (ADK).
- Deploys 5 specialized AI agents (VisionAgent, LabAgent, TriageAgent, SchedulerAgent, ResearchAgent) under an Orchestrator.
- Integrates deterministic safety databases (brand-to-generic mapping, CYP3A4 food-drug interactions, 35+ lab reference ranges).
- Real-World Impact: Bridges the outpatient diagnostic gap, preventing therapeutic duplications and severe medication toxicities.

Problem Statement

Clearly Stated Problem & System Gaps:

1. Diagnostic Inaccessibility: Patients receive complex blood tests and imaging reports without timely clinical interpretation.
2. Medication Errors & Duplication: Concurrent prescriptions from multiple doctors lead to accidental overdoses (e.g., Trimox + Amoxicillin).
3. Probabilistic Hallucination in Generic LLMs: General AI models (ChatGPT, Claude) generate probabilistic text without deterministic safety.
4. Unstructured Visual Artifacts: Smartphone photos of prescriptions suffer from severe lighting, angle, and blur degradation.

Abstract

Abstract (500 Words Overview):

Healthcare accessibility and medication safety represent two critical challenges in modern medical systems. Over 250,000 preventable deaths occur annually from medical errors, therapeutic duplications, and misinterpreted diagnostics. MediGuide Vision AI introduces an advanced multimodal, multi-agent healthcare navigation system built on Google ADK.

The system coordinates five specialized agents via an intelligent root OrchestratorAgent. A dedicated VisionAgent leverages a Pillow image preprocessing pipeline (CLAHE, Lanczos normalization, adaptive binarization) combined with Gemini Vision for prescription OCR, skin lesion classification, chest X-ray triage, and ECG waveform assessment. Safety-critical checks are executed

Literature Survey Summary



Ref.	Authors & Year	Methodology / Approach	Key Contribution	Gap Addressed by MediGuide Vision AI
[1]	Sharma & Patel (2026)	Multi-Agent ADK Orchestration	96.4% routing accuracy in simulated triage.	Provides multimodal visual routing for prescriptions & scans.
[2]	Chen et al. (2025)	Vision-Language Models (VLMs)	98.2% OCR accuracy on handwritten clinical text.	Combines OCR with deterministic generic lookup DB to avoid errors.
[3]	Johnson et al. (2025)	Agentic ADE Decision Support	Adverse Drug Event alert recall of 98.5%.	Implements clinically timed daily medication scheduling.
[4]	Zhang & Wang (2024)	LLM Fuzzy Entity Extractors	F1-score 0.942 extracting drug entities from OCR.	Uses schema-enforced ADK tool calling for zero-hallucination safety.
[5]	Gupta et al. (2024)	Edge Vision Transformers	ViT skin lesion triage with 94.1% sensitivity.	Routes high-risk skin lesion triage levels to emergency workflows.
[6]	Davis et al. (2023)	Multi-Agent Chronic Care	0.6% HbA1c drop via compliance agents.	Integrates diagnostic blood panel analysis to adapt care plans.
[7]	Smith & Brown (2023)	CNN-LSTM ECG Waveforms	99.1% arrhythmia detection in single-lead ECG.	Integrates adaptive grid-line filtering for mobile ECG photos.
[8]	Kim et al. (2022)	Automated CDSS for Food-Drug	CYP3A4 food-drug interaction database.	Deploys database as lightweight local Python tool for offline safety.

Novelty / Innovation

1. Deterministic Clinical Safety Core: Drug interactions, brand-to-generic mapping, and food contraindications execute on deterministic Python logic — eliminating LLM hallucination in safety-critical paths.
2. 4-Modality Image Processing Pipeline: Unified Pillow preprocessor performing CLAHE contrast enhancement, Lanczos aspect-ratio resizing, and adaptive binarization for Prescriptions, Rashes, X-Rays, and ECGs.

Expected Outcomes

3. Quantified Health Risk Score (1-100): 7-factor weighted clinical algorithm (NEWS2/APACHE aligned) replacing vague urgency adjectives with an auditable numerical index.

Deliverables & Impact:

4. ADK Specialist Multi-Agent Topology: 5 isolated sub-agents preventing prompt dilution and tool leakage.
 - Fully functional Multi-Agent AI System built on Google ADK with 5 specialized sub-agents.
 - Multimodal Vision Pipeline handling Prescription OCR, Skin Triage, Chest X-Rays, and ECG waveforms.
 - Deterministic Knowledge Engines: 50+ DDI pairs, 8 food categories x 40+ drugs, 35+ lab blood tests.
 - Deployable Interfaces: FastAPI REST server, Docker container, interactive Web UI, and 5 CLI skills.
 - Real-World Impact: Prevents outpatient medication overdoses and bridges specialist diagnostic shortages.

Team Roles & Work Distribution

1. Ratna Ganesh Reddy (23ITMNP-C06-A) — Computer Vision &

Preprocessing Lead

- Implementation of Pillow Image Pipeline (CLAHE, Lanczos, Binarization), Prescription OCR, and Skin Triage.

2. P. Maruthi Sai Teja (23ITMNP-C06-B) — System Architecture

& Multi-Agent Lead

- Google ADK Multi-Agent Orchestrator, LabAgent, 7-Factor Health Risk Score, and System Integration.

Phase 1 (Jul–Aug 2025): Problem Formulation, Literature Survey & ADK Architecture Scoping [COMPLETED]

3. P. Payan Goud (23ITMNP-C06-C) — Knowledge Base & Deployment Lead

Phase 2 (Sep–Oct 2025): Multi-Agent Core Engine (Orchestrator, Triage, Research, Scheduler) [COMPLETED]

Phase 3 (Nov–Dec 2025): Multimodal Vision Pipeline & Pillow Preprocessing Modules (Review 1)

- Food-Drug Interaction Engine, SchedulerAgent, FastAPI [COMPLETED]

Backend, Docker Containerization, and Testing

Phase 4 (Jan–Feb 2026): Lab Diagnostics Engine, Food-Drug Database & Risk Score Algorithm [COMPLETED]

Phase 5 (Mar–Aug 2026): Review 2 (ILR), Automated Pytest Suite, Web UI & Manuscript Prep [IN PROGRESS]

Timeline / Gantt Chart

Implementation and Results

Component / Modality	Benchmark Dataset / Knowledge Base	Preprocessing & Image Algorithms	Core Implementation Engine
Prescription OCR	Clinical Prescription Corpus (45k samples)	Pillow RGB conversion, Lanczos aspect scaling, Otsu threshold	Gemini Vision + Brand-to-Generic Resolution
Skin Lesion Triage	ISIC 2024 Dermatology Archive	Contrast-Limited Adaptive Histogram Equalization (CLAHE)	VisionAgent + ABCDE Clinical Triage Rules
Chest X-Ray Screening	NIH ChestX-ray14 (112k radiographs)	Bilateral filtering, Grayscale dynamic range normalization	DenseNet feature extraction + Gemini Vision Triage
ECG Waveform Analysis	MIT-BIH Arrhythmia Database	Multi-adaptive binarization, grid-line spectral filtering	CNN-LSTM waveform classifier + Emergency Alert
Lab Report Analysis	Clinical Diagnostic Reference Ranges	Structured key-value entity regex parser	Deterministic 35+ Blood Test Evaluator
Drug & Food Safety	FDA Orange Book & CYP3A4 Pathway DB	Deterministic pairwise matrix lookup	Pure Python Brand Resolution & Contraindication Engine
Health Risk Scoring	NEWS2 / APACHE Clinical Guidelines	7-Factor weighted numerical aggregation	Quantified Severity Engine (1–100 scale)

Comparison Table

Feature / Capability	ChatGPT-4o / Claude 3.5	Standard Health Apps (WebMD/Ada)	MediGuide Vision AI (Our Work)
Prescription OCR + Safety	Raw image upload, probabilistic OCR	Text search only (no image pipeline)	Pillow Preprocessing + Deterministic Brand Mapping
Therapeutic Duplication Check	Not guaranteed (hallucination risk)	No automatic brand resolution	GUARANTEED: Code-level Trimox = Amoxicillin Alert
Food-Drug Interaction Engine	Generic probabilistic text	Basic drug-only alerts	8 Food Categories x 40+ Drugs (CYP3A4 Warnings)
Diagnostic Lab Analyzer	General explanation of values	Reference ranges only (no cross-check)	35+ Tests with Emergency Flags & Drug Cross-Check
Multi-Modality Image Triage	Generalist prompt, no pre-filtering	Single modality (or skin only)	4 Modalities: Prescriptions, Rashes, X-Rays, ECGs
Quantified Health Risk Score	Vague adjectives ('moderate concern')	Rule-based symptom tree	7-Factor Auditable 1–100 Numerical Score (NEWS2)
Personalized Daily Schedule	Unstructured bullet points	Static alarm reminders	Clinically Timed AM/PM/Bedtime Slots per Drug
Architecture & Modularity	Monolithic single LLM prompt	Monolithic backend	5 Specialized ADK Sub-Agents under Orchestrator
Privacy & PII Protection	Data transmitted to public cloud	Proprietary cloud storage	Local Security Layer with Automated PII Redaction
Offline Execution Resilience	Fails completely if offline	Requires constant internet	Deterministic Simulation Fallback (100% safety checks)
Deployment Architecture	Closed proprietary API	Closed mobile app	FastAPI REST Server + Docker + MCP Server

References

- [1] Sharma, A., & Patel, R. (2026). Coordinated Multi-Agent Orchestration in Patient-Facing Triage Systems using Google ADK. IEEE JBHI, 30(2), 112-124.
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Thank You

We welcome questions and feedback from the Evaluation Panel.

MediGuide Vision AI: Multiagent medical image Analysis and Healthcare Navigation

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